

Integrating Android with *jQuery mobile*



The Android market is growing steadily. Every day, new technologies and new frameworks are being introduced. Currently, the application's beauty, creativity and interactivity are as important as its functionality in order for it to provide the best user experience. Among the several Android development platforms, jQuery Mobile can make your application amazingly beautiful. With a user interface system that works seamlessly across all mobile device platforms, it provides a very feature-rich development environment. This article demonstrates the use of jQuery Mobile with PhoneGap.

My system has the following software: the Linux distro is the Ubuntu 10.10 32-bit desktop version; I use Eclipse Indigo, and my Android version is 2.2 (though this can be applied to 2.3 also). The PhoneGap version is 0.9.3 and jQuery Mobile is version 1.2.0. The information and images in this article are with respect to this configuration.

Getting started

The framework provides a set of touch-friendly UI widgets and includes list-views, buttons, form elements, transitions, themes, pages, dialogue boxes, toolbars and an AJAX-powered navigation system to support animated page transitions. To get started with jQuery Mobile, let's try a simple HTML page. In your text-editor, enter the Page-Template (*index.html*) given below, save it and open in a browser:

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
  <title>My Page</title>
  <meta name="viewport" content="width=device-width, initial-
scale=1">
  <link rel="stylesheet" href="jquery.mobile-1.2.0.min.css"/>
  <script src="jquery-1.8.2.min.js"></script>
  <script src="jquery.mobile-1.2.0.min.js"></script>
</head>
<body>
<div data-role="page">
  <div data-role="header">
    <h1>My Title</h1>
  </div><!-- /header -->
  <div data-role="content">
    <p>Hello Phonegap</p>
```

```
</div><!-- /content -->
</div><!-- /page -->
</body>
</html>
```

To help newbies understand the template, let me explain a little more in detail. In the *head*, a *meta viewport* tag sets the screen width to the pixel width of the device. The stylesheet is added since it is needed to implement jQuery effects. We reference the jQuery and jQuery Mobile scripts to add their functionality. As a normal page, it contains a header, body and footer. Here, the *data-role="page"* wrapper signifies that a page is going to be delineated. The *data-role="header"* signifies a header tag, while *data-role="content"* signifies a body for the whole content. Inside this content container, you can add any standard HTML elements—headings, lists, paragraphs, etc.

Let's proceed to build, implement and run this jQuery-enabled HTML page in the Android emulator.

Building and configuring

The first thing is to download and install the Android SDK—the software development kit that provides the necessary tools for development, and comes in different versions for Windows, Linux and Mac OS. Then we need to install an IDE—I use Eclipse. Download and install Eclipse, then launch it and open its workbench. Download and install the ADT plug-in for Eclipse. Next, download the PhoneGap package—the cross-platform framework used to create package files to install your application on different mobile OSs like Android, iOS, Blackberry, etc. Also download the jQuery Mobile pre-defined *min.js* and *min.css* files used in our *index.html* page.

Now that we have all the ingredients, we can start with integration. Start Eclipse and choose your workspace. Go to *File > New > Android Project*. Specify a *Project Name* (e.g., *My First*